

Parallel Multimodal Circuits Control an Innate Foraging Behavior

Highlights

- The metabotropic glutamate receptor MGL-1 is necessary for local search behavior
- Parallel chemosensory and mechanosensory circuits redundantly generate local search
- Sensory neurons encode information about the time since the last food encounter
- The motor circuit for local search becomes resistant to sensory input over time

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In Brief

Animals engage in local search behavior after recent encounters with food-related sensory cues. López-Cruz et al. show that parallel chemosensory or mechanosensory circuits can independently trigger local search after food removal in *C. elegans*, each using a similar glutamate receptor module.



Parallel Multimodal Circuits Control an Innate Foraging Behavior

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SUMMARY

Foraging strategies emerge from genetically encoded programs that are similar across animal species. Here, we examine circuits that control a conserved foraging state, local search behavior after food removal, in *Caenorhabditis elegans*. We show that local search is triggered by two parallel groups of chemosensory and mechanosensory glutamatergic neurons that detect food-related cues. Each group of sensory neurons suppresses distinct integrating neurons through a G protein-coupled metabotropic glutamate receptor, MGL-1, to release local search. The chemosensory and mechanosensory modules are separate and redundant; glutamate release from either module can drive the full behavior. A transition from local search to global search over several minutes after food removal is associated with two changes in circuit function. First, the spontaneous activity of sensory neurons falls. Second, the motor pattern generator for local search becomes less responsive to sensory input. This multimodal, distributed short-term food memory provides robust control of an innate behavior.

INTRODUCTION

The optimal foraging strategy across conditions and species can be formulated in a unified framework combining resource distribution, recent experience, and current internal state (Humphries et al., 2010; Nathan et al., 2008; Salvador et al., 2014; Weimerskirch et al., 2007). One prominent foraging strategy within this framework is area-restricted or local search, an intensive exploration over several minutes of the region where food resources were last encountered. As time since the last food encounter passes, animals transition to global search strategies to explore distant areas. The local-to-global search foraging pattern has been observed in fish (Papastamatiou et al., 2012), reptiles (Eifler et al., 2012), insects (Bell, 1985; Lihoreau et al., 2016; Nakamuta, 1985), birds (Dias et al., 2009; Weimerskirch et al., 2007), and

mammals (Benedix, 1993; Hills et al., 2013). Although the motor programs employed during local search vary greatly between these species—swimming, crawling, flying, walking—the underlying behavioral state consisting of a sustained period of intensive searching after food encounter is highly conserved. Understanding the molecular and circuit mechanisms underlying this searching state may provide insight into the basis of ancient conserved behaviors.

Two features of local search hint at circuit elements that may be required for search behavior. First, animals execute local search not only after recent food encounters, but also after encounters with sensory cues associated with resources, such as food odors (Bell, 1985; Murata et al., 2017), light levels (Horstick et al., 2017), and pheromones (Nakamuta, 1985). This observation suggests that local search involves a neuronal representation of food-related sensory features. Second, search behavior is persistent and outlasts its trigger (previous food encounter), suggesting that it results from an internally generated short-term food memory. While the external triggers and dynamics of local search have been extensively characterized, the neuronal circuits that represent the external sensory environment and the sites that hold the short-term food memory are not fully understood.

The compact nervous system of *C. elegans*, which consists of 302 neurons, presents an opportunity to explore these questions. *C. elegans* feeds on bacterial food (Brenner, 1974) and many neurons that detect food-related sensory cues have been characterized. For example, the AWC olfactory neurons detect volatile bacterial odors, ASK neurons detect amino acids, and several mechanosensory neurons detect food texture (Albeg et al., 2011; Bargmann, 2006; Park et al., 2008; Sawin et al., 2000; Yemini et al., 2013). The synaptic connections between these sensory neurons and the downstream interneurons and motor neurons that generate behaviors are known (White et al., 1986), making this an excellent system for defining neuronal circuits related to foraging.

Upon removal from food, *C. elegans* executes a stereotypical local search behavior in which it explores a small area by executing many random, undirected reversals and turns for ~15 min. If it fails to find food, it transitions to a global search in which it explores larger areas by suppressing reorientations and executing long forward runs (Ahmadi and Roy, 2016; Calhoun et al., 2014; Gray et al., 2005; Hills et al., 2004; Salvador



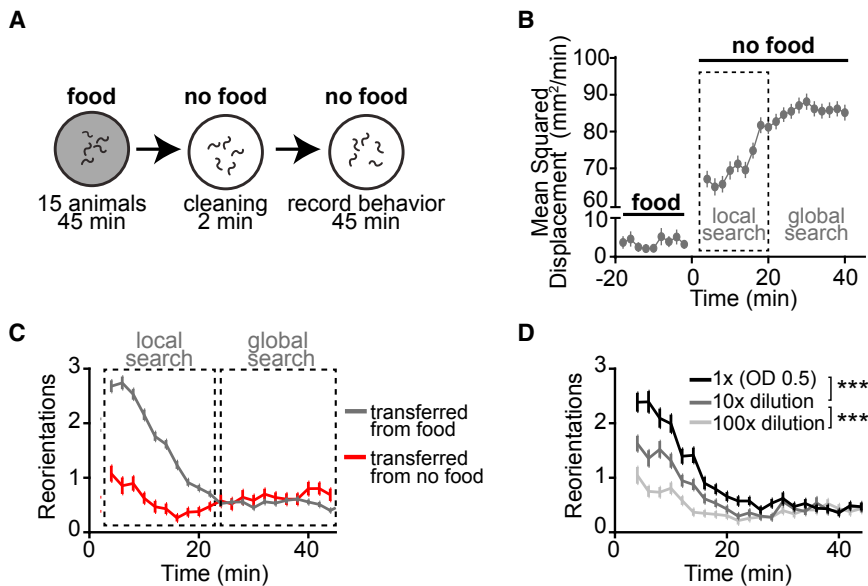


Figure 1. Off-Food Foraging in Wild-Type Animals

(A) Schematic illustration of off-food foraging assay. Animals were transferred from a homogeneous bacterial food lawn to a large plate (80 cm²) without food, where their behavior was filmed for 45 min. The behavior varies from day to day, so all experiments were compared to control animals recorded simultaneously in an adjacent behavioral chamber. (B) Average mean squared displacement per minute on food and after food removal. (C) Gray line: off-food foraging showing mean reorientations in 2-min time windows after food removal. The first 2 min are not plotted as these are affected by transfer (Zhao et al., 2003). Red line, foraging behavior of animals pre-incubated for 45 min on a plate without food. (D) Effect of food concentration on subsequent off-food foraging. Animals were placed on either a standard OD = 0.5 bacterial lawn (n = 103), a lawn diluted 10-fold (n = 114), or a lawn diluted 100-fold (n = 100), removed after 45 min, and their off-food behavior was recorded. All data are presented as mean ± SEM. ***p < 0.001 by Wilcoxon rank-sum test with Bonferroni correction.

et al., 2014; Wakabayashi et al., 2004). The reorientations associated with local search in *C. elegans* are present in different patterns in other behaviors such as chemotaxis (Pierce-Shimomura et al., 1999) and aerotaxis (Hums et al., 2016). Previous studies have identified sensory neurons that contribute to acute reorientations during local search by activating glutamate-gated ion channels on downstream interneurons (Chalasani et al., 2007; Gray et al., 2005; Hills et al., 2004; Wakabayashi et al., 2004). However, ion channels signal over sub-second timescales, whereas local search occurs over minutes. The relationship between acute sensory signals and the longer-term dynamics of the local search state is not known.

Here, we identify a G protein-coupled glutamate receptor, MGL-1, as a transducer of sensory information into sustained local search. Glutamatergic chemosensory and mechanosensory neurons generate local search by suppressing MGL-1-expressing neurons, whose activity inhibits the local search state. The food memory can be represented by either the chemosensory or the mechanosensory circuit, consistent with the multimodal nature of food cues, and fades through a combination of decreased sensory activity and decreased sensorimotor coupling. Our findings reveal a circuit organization that ensures robust execution of a conserved adaptive behavior.

RESULTS

Reorientation Patterns and Food History in Local Search Behavior

We studied local search behavior by transferring animals from a standard bacterial food lawn to a large agar plate without food (~80 cm²), where we filmed and quantified their behavior continuously for 45 min (Figure 1A). As described previously (Calhoun et al., 2014; Gray et al., 2005; Hills et al., 2004; Wakabayashi et al., 2004), animals initially explored a small area by performing

many reorientations (Figures 1B and 1C, local search), then gradually transitioned to exploring a larger area by performing fewer reorientations over time (Figures 1B and 1C, global search). Following precedent, we defined local search as the period over which recent food exposure altered reorientation frequencies, and global search as the lower-reorientation state that did not vary with food history (Figure 1C). *C. elegans* can perform a variety of reorientations using different locomotor sequences (Figure S1). We found that reorientations that start with a reversal were increased during local search (time-dependent, Figures S1A–S1C), but reorientations resulting from turning during forward movement occurred at similar frequencies during local and global search (time-independent; Figures S1D and S1E). Here, we focused on the time-dependent reorientations that drive local search behavior (Figure 1C).

If local search represents a short-term memory of recent food experience, it should be modulated by food history. To test this prediction, we varied the food concentration that animals experienced for 45 min prior to food removal. Animals that had been in dilute food performed fewer reorientations during local search than animals that had been in concentrated food (Figure 1D). Forward locomotion speed and reorientations during global search were unchanged by prior food concentrations (Figures 1D and S1F). This result and others (Calhoun et al., 2015), suggest that local search is a memory state that is actively generated and modulated in response to food experience and removal and that it transitions over time to a default global search state.

The transition from local to global search is gradual at a population level (Figure 1C). To characterize the transition in individuals, we looked at reorientation dynamics for 1,631 single animals and searched for abrupt transitions in reorientation frequency (Figures S2A–S2C). 51% of animals appeared to have a single abrupt transition corresponding to the end of local search, whereas 49% had multiple transitions or no clear

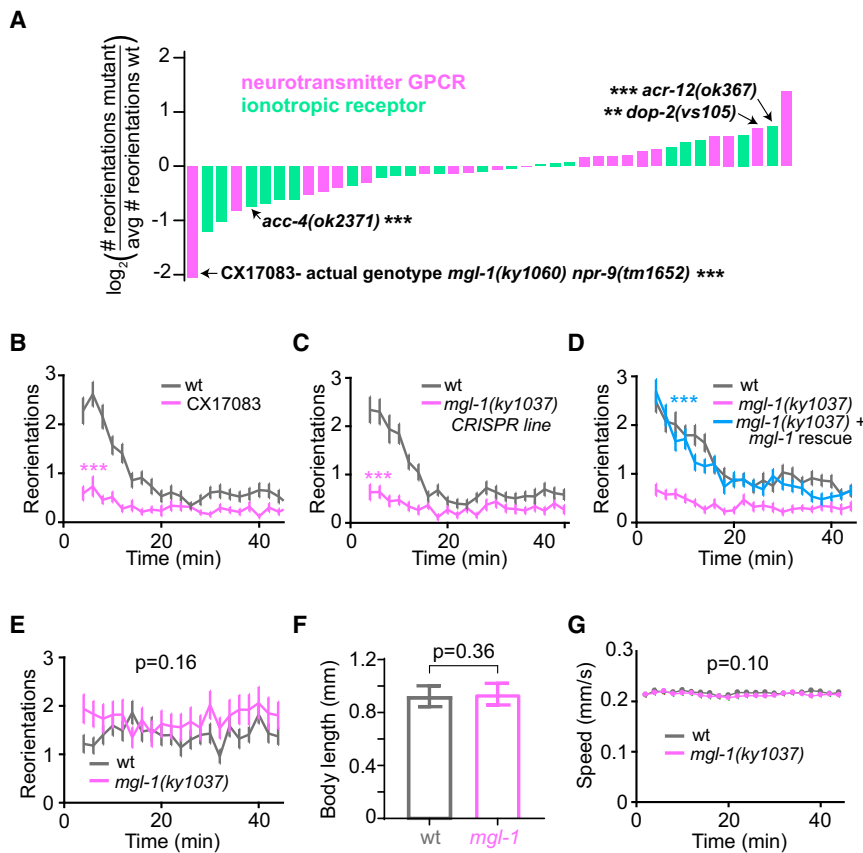


Figure 2. *mgl-1*, a Metabotropic Glutamate Receptor, Is Necessary for Local Search

(A) Local search (2–20 min) reorientation rates normalized to wild-type controls for 42 mutant strains. Data show the fold change in reorientations during local search for each mutant strain, presented as the log₂ of mutant/wild-type reorientation ratio. We focused on mutants that met two criteria: (1) showed a statistically significant fold change during local search of at least 1.5 ($\text{abs}(\log_2(\text{ratio})) > 0.58$), and (2) local search was affected more than global search. For more information, see Table S1.

(B) Off-food foraging for mutant strain *CX17083* [*npr-9(tm1652) mgl-1(ky1060)*] (wild-type [WT], n = 56; *CX17083*, n = 62).

(C) Off-food foraging for CRISPR-generated *mgl-1(ky1037)* mutant (WT, n = 61; *mgl-1(ky1037)*, n = 41).

(D) The foraging defect in *mgl-1(ky1037)* (pink) is rescued by a genomic fragment encompassing the *mgl-1* genomic region (blue) (*mgl-1(ky1037)*, n = 54; *mgl-1* rescue, n = 66).

(E) Time-independent reorientations in wild-type and *mgl-1(ky1037)* animals (WT, n = 61; *mgl-1(ky1037)*, n = 41; see also Figure S1).

(F) Body length of age-matched young adult wild-type and *mgl-1(ky1037)* animals (WT, n = 61; *mgl-1(ky1037)*, n = 41).

(G) Average forward locomotion speed for wild-type and *mgl-1(ky1037)* animals (WT, n = 61; *mgl-1(ky1037)*, n = 41).

Results in (B)–(E) and (G) are presented as mean $\pm$ SEM. Results in (F) are presented as mean $\pm$ SD. Data from (C) and (E)–(G) are taken from the same experiments. p values calculated using Wilcoxon rank-sum test with Bonferroni correction ***p < 0.001.

transition (Figures S2A and S2B). There was a wide range in the duration and the intensity of local search behaviors in individual animals (Figures S2C and S2D). Despite this variability, 93% of the animals performed more reorientations at early times after food removal (Figure S2E), indicating that local search is a robust behavioral response to recent food experience.

The Metabotropic Glutamate Receptor MGL-1 Is Necessary for Local Search

To gain insight into the neuronal circuit mechanisms that generate local search behavior, we filmed behavior in 42 candidate mutant strains lacking individual ionotropic and metabotropic neurotransmitter receptors expressed in neurons of the foraging circuit, with a focus on mutants that did not cause defects in feeding or general locomotion (Gray et al., 2005; Flavell et al., 2013). For each mutant strain, we calculated the average fold-change in reorientations relative to wild-type during local search (2–20 min) (Figure 2A; Table S1) and global search (30–45 min) (Table S1). After setting a threshold for effect size (1.5-fold), we selected mutants that had statistically stronger defects during local search than during global search. Of the 42 mutant strains tested, the mutant strain *CX17083* showed the strongest defect in local search behavior (Figure 2B; Table S1).

CX17083 was profoundly defective in local search, with milder defects in reorientation rates during global search (Figure 2B; Table S1). It was originally selected for its annotated *npr-9(tm1552)* mutation (Bendena et al., 2008) but we were unable to rescue its local search defect by rescuing functional *npr-9* expression with a transgene (Figure S3A), and a new CRISPR-generated *npr-9* mutant allele did not recapitulate the phenotype (Figure S3B), suggesting that a second background mutation caused the observed phenotype. To identify the unmapped background mutation, we sequenced *CX17083* and identified a deletion that disrupted several exons of *mgl-1*, which encodes a G-protein coupled inhibitory metabotropic glutamate receptor (Figure S3C) (Dillon et al., 2006, 2015). A full-length *mgl-1* transgene rescued local search in *CX17083*, implicating *mgl-1* as the causative mutation (Figure S3D).

To strengthen the identification of *mgl-1*, we generated a new *mgl-1* mutant allele using CRISPR/Cas9 (see STAR Methods). This *mgl-1(ky1037)* strain recapitulated the *CX17083* local search defect (Figure 2C) and was rescued by a *mgl-1* transgene (Figure 2D), confirming that *mgl-1* supports local search. *mgl-1* mutants had normal time-independent reorientations (Figure 2E), normal adult body length (Figure 2F), and normal forward locomotion speed (Figure 2G). *mgl-1* has been previously demonstrated to have persistent pharyngeal pumping after food

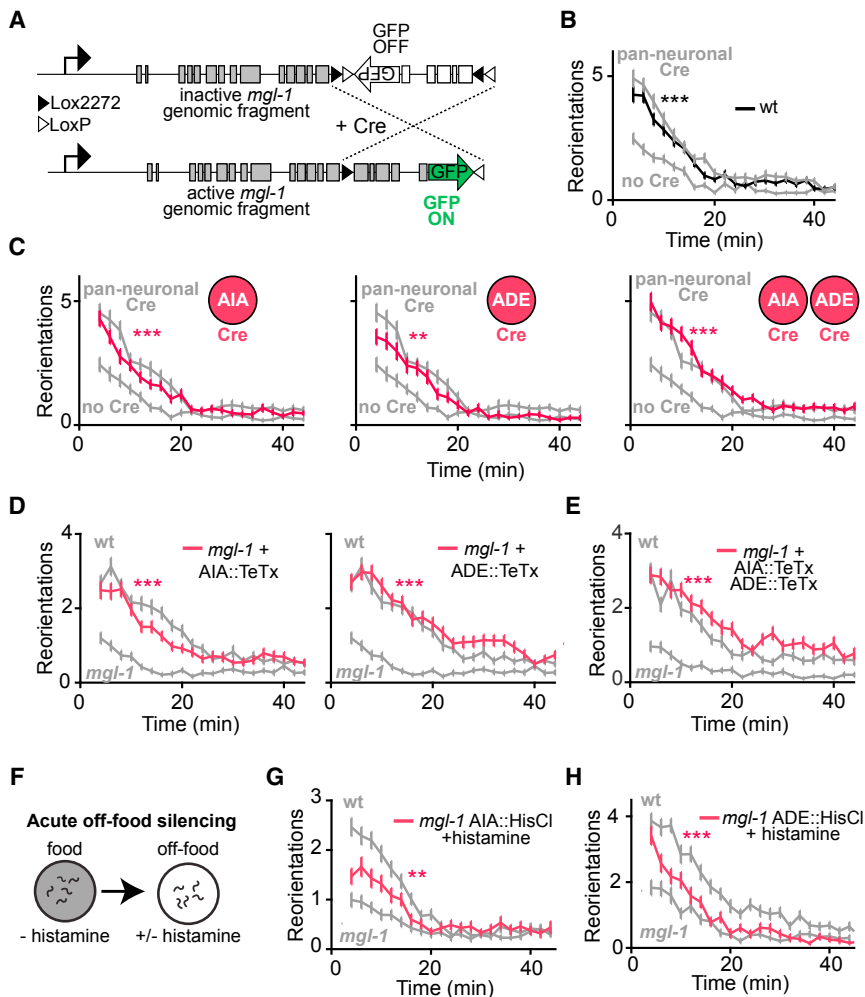


Figure 3. *mgl-1* Suppresses AIA and ADE to Generate Local Search

(A–C) *mgl-1* rescue in subsets of its endogenous expression pattern using an intersectional transgene approach. (A) Schematic depicting Cre-Lox transgene approach. Successful intersectional activation is confirmed by GFP, which is expressed with *mgl-1* in a bicistronic transcript (e.g., Figure S4B). (B) Rescue of *mgl-1* by reconstituting inverted transgene in all neurons using pan-neuronal Cre (*tag-168::nCre*). Animals that only express the inverted transgene are labeled “no Cre” (no Cre, n = 72; pan-neuronal Cre, n = 69). (C) Rescue of *mgl-1* by expression in AIA, ADE or both neurons (no Cre, n = 46; AIA rescue, n = 50; ADE rescue, n = 32; AIA + ADE rescue, n = 48). Baseline reorientation rates during local search vary across days and were high during this block of experiments, but controls conducted across different days confirm that the inverted transgene alone does not rescue *mgl-1* (Figure S4A).

(D) Rescue of *mgl-1(ky1037)* phenotype by expression of tetanus toxin light chain (TeTx) in AIA (and NSM) or ADE (and CEP and PDE) neurons (*mgl-1*, n = 78; AIA::TeTx *mgl-1*, n = 63; ADE::TeTx *mgl-1*, n = 82). Control strains implicate AIA and ADE as the relevant sites of expression (Figures S4F and S4H and legend).

(E) Rescue of *mgl-1(ky1037)* phenotype by expression of tetanus toxin light chain (TeTx) in both AIA and ADE (as well as NSM, CEP, and PDE) (*mgl-1*, n = 72; AIA::TeTx + ADE::TeTx *mgl-1*, n = 69).

(F) Acute neuronal silencing by expressing cell-specific histamine-gated chloride channel and adding exogenous histamine to plates. Negative controls are *mgl-1(ky1037)* animals expressing the histamine-gated chloride channel that were transferred to plates without histamine.

(G) Acutely silencing AIA off food partially rescues *mgl-1(ky1037)* local search (*mgl-1* AIA::HisCl – his, n = 84; *mgl-1* AIA::HisCl + his, n = 89).

(H) Acutely silencing ADE (and PDE in a fraction of animals) off food partially rescues *mgl-1* local search (*mgl-1* ADE::HisCl – his, n = 60; *mgl-1* ADE::HisCl + his, n = 51).

All data presented as mean ± SEM. p values calculated using Wilcoxon rank-sum test with Bonferroni correction. **p < 0.01 ***p < 0.001.

removal and altered physiological and behavioral responses to prolonged starvation (Greer et al., 2008; Kang and Avery, 2009a, 2009b; Dillon et al., 2015; Ahmadi and Roy, 2016). Our results indicate that *mgl-1* is necessary for local search behavior after removal from food.

mgl-1 Suppresses AIA and ADE Neurons to Generate Local Search

To identify the circuits where *mgl-1* acts to generate local search, we rescued *mgl-1* in subsets of its endogenous expression pattern using an intersectional transgene approach (Figure 3A). A transgene containing an inverted, inactive floxed *mgl-1* genomic fragment was activated in a subset of cells by a second transgene expressing Cre recombinase under cell-specific promoters. Successful recombination was confirmed by GFP expression. In control experiments, the inactive fragment alone had minimal effects on local search behavior in *mgl-1* mutants (Figure S4A). Activating the *mgl-1* fragment in all neurons rescued the *mgl-1* local search defect (Figure 3B) and resulted

in broad GFP expression (data not shown). To narrow down the sites of *mgl-1* action, we activated the *mgl-1* fragment in smaller subsets of cells. Simultaneous activation in sensory neurons ASI, IL1, and ADE, the interneuron AIA, and the motor neurons RMD and NSM rescued local search (Figure S4B). Of these neurons, activating *mgl-1* in AIA and ADE or both neurons (Figure 3C), but not in RMD, ASI, NSM, or IL1 (Figure S4C), was sufficient for rescue. In control experiments, transgenes with nonsense mutations in the *mgl-1* coding region failed to rescue the behavior (Figures S4D and S4E). *mgl-1* had been previously reported to be expressed in AIA, but not in ADE (Greer et al., 2008). The expanded expression we observed is likely due to the use of a genomic fragment that includes all *mgl-1* introns.

MGL-1 is most similar in sequence to mammalian group II metabotropic glutamate receptors, which couple to inhibitory G_o or G_i proteins (Dillon et al., 2015). By analogy, a loss of inhibition in *mgl-1* mutants might result in hyperactive AIA and ADE neurons that interfere with local search. To test this possibility, we expressed tetanus toxin light chain in AIA, ADE, or both neurons

in *mgl-1* mutant animals (Figures 3D, 3E, and S4F). Inhibiting neurotransmitter release from either AIA, ADE, or from both neurons with tetanus toxin rescued local search behavior in *mgl-1* mutants, with minimal effects on locomotion speed (Figures 3D, 3E, and S4F–S4H). Expressing tetanus toxin in AIA and ADE in wild-type animals did not affect search behavior (Figure S4I). These results suggest that inhibiting synaptic release with tetanus toxin can substitute for *mgl-1* action to restore local search.

Neurons expressing tetanus toxin lack synaptic vesicle release chronically throughout life. To silence AIA or ADE acutely during foraging behavior, we expressed the *Drosophila* histamine-gated chloride channel HisCl1 in either neuron (Pokala et al., 2014). *C. elegans* does not use histamine as an endogenous transmitter, so selective expression of HisCl1 permits rapid silencing of target neurons by adding histamine to the assay plate (Figure 3F). Acutely silencing AIA or ADE off food partially rescued the local search defect in *mgl-1* mutants, albeit to a lesser extent than tetanus toxin (Figures 3G and 3H) (see Discussion).

Together, these results indicate that when animals are removed from food, MGL-1 suppresses AIA and ADE to release local search behavior. This interpretation in turn suggests that AIA and ADE can suppress local search, and indeed an inhibitory effect of AIA on reorientations has been reported (Wakabayashi et al., 2004). AIA and ADE each release multiple neurotransmitters and peptides that could mediate this behavioral effect. We tested two candidates, dopamine from ADE and *ins-1* from AIA, but found that neither explains local search behavior (Figure S4J) (see Discussion).

Parallel, Multimodal Glutamatergic Sensory Pathways Redundantly Control Foraging Behavior

We next focused on the sources of glutamate that suppress AIA and ADE through MGL-1 to generate local search. AIA and ADE do not express *eat-4*, the transporter that loads glutamate into synaptic vesicles (Serrano-Saiz et al., 2013), but they are post-synaptic to multiple glutamatergic neurons. The wiring diagram of *C. elegans* (White et al., 1986) and *eat-4* expression data (Serrano-Saiz et al., 2013) predict that AIA receives glutamatergic input from several chemosensory neurons that detect food-related volatile odors and soluble chemicals (Chalasani et al., 2007; Wakabayashi et al., 2009) (Figure 7). ADE is predicted to receive glutamatergic input from mechanosensory neurons regulated by food texture (Albeg et al., 2011; Serrano-Saiz et al., 2013; White et al., 1986; Yemini et al., 2013) (Figure 7).

To characterize the role of sensory glutamate inputs in local search behavior, we modified the endogenous *eat-4* (VGLUT1) locus of wild-type animals to permit cell-specific knockout of glutamate release. Two successive rounds of CRISPR/Cas9 editing were performed to insert an FRT before the start codon and *let-858-3'-UTR::FRT::mCherry* after the stop codon of the endogenous *eat-4* coding region (Figure 4A) (Schwartz and Jorgensen, 2016). In this edited *eat-4* strain, expression of a nuclear localized flippase (nFLP) under cell-specific promoters excised the *eat-4* coding region and resulted in the appearance of mCherry in the targeted cells (Figure 4A; see STAR Methods). To validate this approach, we expressed nFLP under a pan-

neuronal promoter (*tag-168::nFLP*); this resulted in a strong reorientation defect reminiscent of *eat-4* mutants (Hills et al., 2004) (Figure 4B) and broad neuronal mCherry expression, consistent with *eat-4* expression patterns (Figures S5A and S5B).

We began by knocking out glutamate release from the individual chemosensory neurons AWC, ASK, and ASG, which respond to the removal of chemosensory cues related to food (Chalasani et al., 2007; Wakabayashi et al., 2009). None of these single cell glutamate knockouts resulted in a striking behavioral defect (Figures 4C–4E). Next, we knocked out glutamate release more broadly: from all chemosensory glutamate neurons that synapse onto AIA, from all mechanosensory glutamate neurons that synapse onto ADE, or from both pathways simultaneously. Knocking out glutamate release from either sensory pathway individually had little effect on local search (Figure 4F, left and middle). However, glutamate knockout from both the chemosensory and mechanosensory pathways resulted in a local search defect that was indistinguishable from that of *mgl-1* mutants (Figure 4F, right, control in Figure S5C).

To relate acute sensory activity to local search behavior, we transiently silenced the relevant glutamatergic sensory neurons of animals performing local search using optogenetic activation of the light-gated chloride channel GtACR2 (Govorunova et al., 2015). GtACR2 was expressed in the glutamatergic chemosensory and mechanosensory neurons that synapse onto AIA and ADE (Figure 7) using an inverted Cre-Lox recombination strategy (see STAR Methods). Silencing the neurons for 30 s with light led to a nearly 2-fold decrease in reorientation probability, indicating that sensory activity is necessary to maintain a high reorientation state during local search (Figure 4G).

The local search defect after chemosensory and mechanosensory glutamate knockout might be due to loss of glutamate signaling to AIA or ADE, or to loss of glutamate signaling to other neurons. To evaluate these possibilities, we inhibited neurotransmitter release from AIA or ADE with tetanus toxin in animals lacking glutamate in both sensory pathways. Tetanus toxin expression in either AIA or ADE fully restored local search behavior (Figure 4H), supporting the importance of AIA and ADE as targets of sensory glutamate release. Taken together, these results suggest that glutamatergic chemosensory and mechanosensory neurons can each independently drive full local search behavior by suppressing either AIA or ADE.

Spontaneous Activity in ASK and AIA Reports Time after Food Removal

Acute food removal leads to a transient activation of multiple glutamatergic sensory neurons (Chalasani et al., 2007; Wakabayashi et al., 2009), but the evolution of neuronal activity over the longer duration of the foraging assay has not been characterized. To investigate neuronal responses to chronic food removal, we monitored spontaneous calcium activity in animals expressing genetically encoded calcium indicators in the glutamatergic ASK sensory neurons and *mgl-1*-expressing AIA interneurons (Akerboom et al., 2012) (Figure 7A). Of the chemosensory neurons, ASK was selected because it was the only neuron to cause a significant, albeit mild, behavioral defect when *eat-4* was knocked out (Figure 4D), and because it is directly activated by food removal (Calhoun et al., 2015; Wakabayashi et al., 2009).

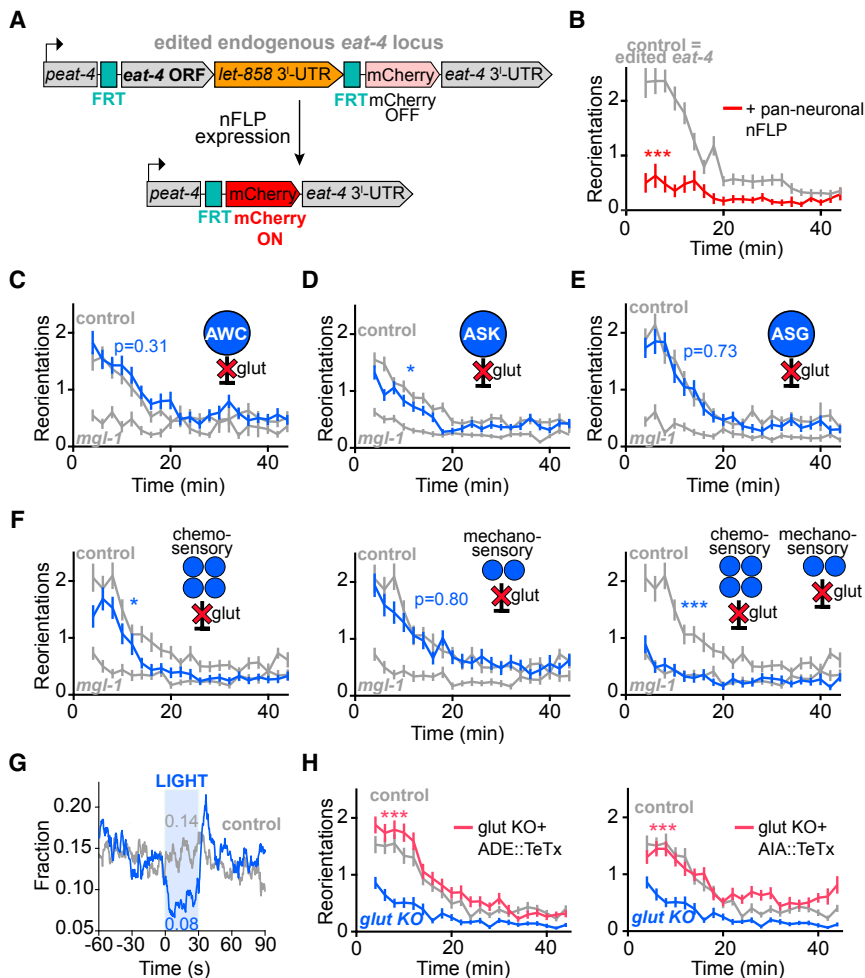


Figure 4. Parallel Glutamatergic Sensory Pathways Drive Local Search

(A) Schematic of endogenous glutamate knockout strategy. Using CRISPR/Cas9, an FRT site was inserted immediately before the start codon of *eat-4* (VGLUT1), and *let-858* 3'-UTR::FRT::mCherry immediately after the stop codon of *eat-4*. *let-858* 3'-UTR stops transcription, so mCherry is not expressed. To knock out glutamate release in this edited strain, nuclear-localized Flippase (nFLP) was expressed under cell-specific promoters, leading to excision of the *eat-4* ORF, confirmed by mCherry expression in the targeted cells.

(B) Validation of glutamate knockout strategy. Pan-neuronal glutamate knock out was achieved by expressing pan-neuronal nFLP (*tag-168::nls-FLP*) in the edited *eat-4* strain. Animals show phenotype reminiscent of *eat-4* mutants (control, *n* = 65; pan-neuronal nFLP, *n* = 56).

(C–E) Off-food foraging after glutamate knockout from individual chemosensory neurons. (C) AWC glutamate knockout (control, *n* = 55; AWC glut KO, *n* = 57). (D) ASK glutamate knockout (control, *n* = 108; ASK glut knockout [KO], *n* = 92). (E) ASG glutamate knockout (control, *n* = 60; ASG glut KO, *n* = 58).

(F) Off-food foraging after glutamate knockout from all *tax-4* chemosensory neurons, all *mec-3* mechanosensory neurons, or both classes (control, *n* = 59; chemosensory glut KO, *n* = 65; mechanosensory glut KO, *n* = 60; chemosensory + mechanosensory glut KO, *n* = 53).

(G) Optogenetic inhibition of local search. Animals expressed the light-activated chloride channel *GtACR2* in glutamatergic chemosensory and mechanosensory neurons that synapse onto AIA and ADE (AWC, ASK, ASE, ASG, FLP, AVM). Controls expressed *GtACR2* and were treated with light, but not pre-incubated with retinal. Data show the instantaneous fraction of animals

performing a reorientation. The mean fraction during stimulation is reported on the plot. Light stimulation was delivered during local search, 2–14 min after food removal (sensory *GtACR2*, *n* = 151; controls, *n* = 159).

(H) Off-food foraging for both mechano- and chemosensory glutamate knockout (glut KO), plus ADE (left) or AIA (right) expression of TeTx (chemo + mechano glut KO, *n* = 77; chemo + mechano glut KO + ADE::TeTx, *n* = 107; chemo + mechano glut KO + AIA::TeTx, *n* = 77).

All data except (G) presented as mean $\pm$ SEM. *p* values calculated using Wilcoxon rank-sum test with Bonferroni correction. **p* < 0.05 ****p* < 0.001.

Animals were removed from food and loaded into a microfluidic device (Chronis et al., 2007), where the spontaneous activities of ASK and AIA were monitored for 10 min immediately after food removal (0–10 min, local search) and again after 40 min (40–50 min, global search) (Figure S6A).

In the absence of food or obvious external stimuli, both ASK and AIA exhibited striking spontaneous activity (Figures 5A and 5B). ASK had spontaneous decreases in calcium levels from a high baseline (negative transients, median length 21 s) (Figures 5A and S6B), whereas AIA had spontaneous increases in calcium levels from a low baseline (positive transients, median length 7 s) (Figures 5B and S6C). These long-duration transients resemble other spontaneous activity states observed in *C. elegans* neurons (Gordus et al., 2015; Skora et al., 2018).

We next compared the activity of ASK and AIA at the early and late times after food removal. At later times after food removal, ASK had more negative transients (Figures 5A and 5C), and AIA had more positive transients (Figures 5B and 5D); the late

AIA transients were also slightly larger in amplitude (Figures 5E, 5F, and 6G). Together, these results indicate that ASK becomes less active and AIA becomes more active during the shift from local to global search behavior.

Glutamate Shapes AIA Activity at Multiple Timescales

Simultaneous calcium imaging from the ASK and AIA neurons revealed that their spontaneous transients often occurred at the same time, such that ASK-negative transients coincided with AIA-positive transients (Figures 6A, 6B, and S7A). This result suggested that glutamate release from ASK might inhibit AIA. Indeed, ASK and AIA transients were only weakly correlated in animals mutant for the vesicular glutamate transporter EAT-4 (Figures 6C, 6D, and S7B), although the number, size, and duration of *eat-4* ASK transients were indistinguishable from wild-type (Figure S7D). The magnitude of AIA transients was also reduced in *eat-4* mutants, consistent with glutamatergic signaling from sensory neurons to AIA (Figures 6H and S7E).

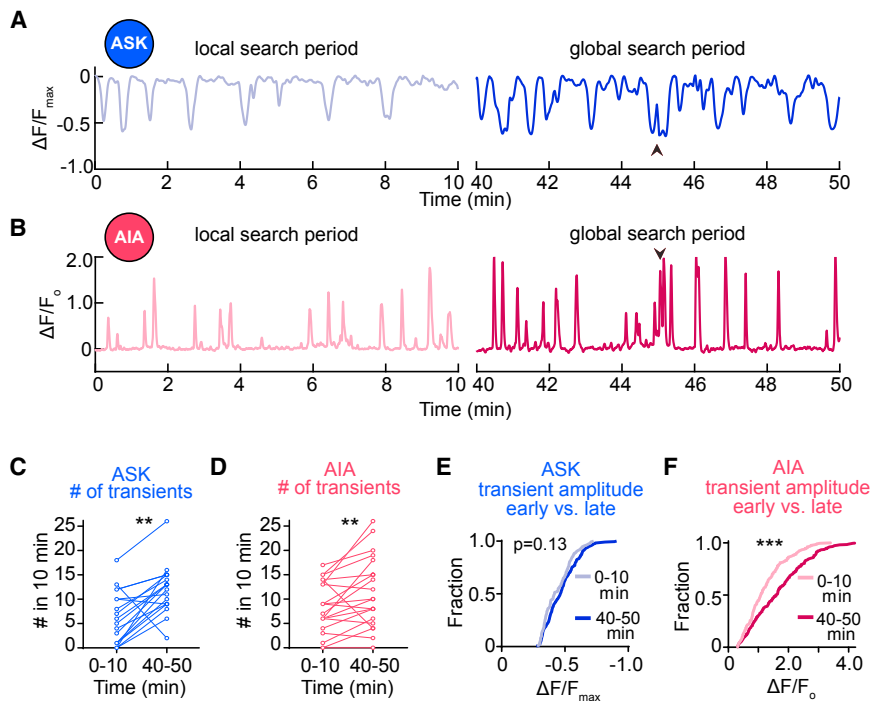


Figure 5. Spontaneous Neuronal Dynamics Represent a Short-Term Food Memory

(A and B) Spontaneous calcium dynamics in a representative ASK neuron (A) or AIA neuron (B) expressing GCaMP5A, recorded 0–10 min and again 40–50 min after food removal.

(C) Number of ASK-negative transients in individual wild-type animals early (0–10 min) versus late (40–50 min) after food removal ($n = 19$). $**p < 0.01$ by Wilcoxon signed-rank test.

(D) Number of AIA-positive transients in individual wild-type animals early (0–10 min) versus late (40–50 min) after food removal ($n = 21$). $**p < 0.01$ by Wilcoxon signed-rank test.

(E) Cumulative ASK-negative transient amplitude early (0–10 min, $n = 111$) versus late (40–50 min, $n = 216$) after food removal for wild-type animals. p value calculated using Kolmogorov-Smirnov test.

(F) Cumulative AIA-positive transient amplitude early (0–10 min, $n = 210$) versus late (40–50 min, $n = 255$) after food removal for wild-type animals. $***p < 0.001$ by Kolmogorov-Smirnov test.

mgl-1 was not required for the temporal coupling of ASK and AIA activity, or the increased number of AIA transients after long times off food (Figures 6E, 6F, S7C, and S7F). *mgl-1* did affect the evolution of activity over time, as AIA calcium transients in *mgl-1* mutants had similar amplitudes at early and late time points after food removal (Figures 6I and S7F). These results suggest that MGL-1 acts together with a second glutamate receptor at the ASK to AIA synapse.

We sought direct evidence for a second glutamate receptor by using whole-cell patch-clamp to record AIA responses to exogenously applied glutamate in dissected animals. 1 mM glutamate pulses under voltage-clamp conditions induced a fast current in AIA (Figure 6J) whose reversal potential matched the equilibrium potential of chloride ions (Figure 6K). This presumptive glutamate-gated chloride current was unaffected in *mgl-1* mutants (Figure 6J).

These results indicate that glutamate signaling through MGL-1 acts alongside glutamate receptors that mediate fast inhibitory synapses between ASK and AIA. Thus, glutamate may act through multiple receptors including MGL-1 to modulate AIA activity at multiple timescales.

The Reorientation Circuit Decreases Its Response to Sensory Input over Time

The results described above demonstrate quantitative changes in ASK activity during the transition from local to global search. To examine associated behavioral changes, we used optogenetic manipulation to assess responses to ASK activation in freely-moving animals. ASK was depolarized in freely-moving animals during local and global search by cell-specific expression of the red-shifted channelrhodopsin variant ReaChR (Lin et al., 2013) and activation by light. Light stimuli were 30 s long

to approximately match the duration of spontaneous high-activity states in ASK (Figure 5A).

Stimulating ASK for 30 s during local search led to a sharp transient increase in reorientations (~ 10 s), a sustained elevation throughout the stimulus, and a post-stimulus suppression of reorientations and return to baseline over 60 s (Figure 7B). During global search, ASK activation also resulted in reorientations, but the peak level of reorientations and the increase in reorientations over baseline were smaller than in local search (Figures 7C and 7D).

These results suggest that ASK is less able to drive reorientations during global search. As a control, we optogenetically activated the RIM interneurons, which belong to a coupled network of neurons downstream of AIA that drives reorientation behavior (Gordus et al., 2015; Kato et al., 2015) (Figure 7A). RIM activation led to equally strong reorientation responses during local search and global search; if anything, responses over baseline were stronger in global search (Figures 7E–7G). The different results of ASK and RIM activation suggest that in addition to changes in ASK activity, the reorientation circuit becomes less sensitive to ASK input during the transition from local search to global search.

DISCUSSION

Foraging patterns incorporating local search have been described in numerous animal species (Bell, 1985; Benedix, 1993; Dias et al., 2009; Eifler et al., 2012; Gray et al., 2005; Hills et al., 2013; Lihoreau et al., 2016; Nakamuta, 1985; Papastamatiou et al., 2012; Wakabayashi et al., 2004; Weimerskirch et al., 2007). In each case, the animal performs a time-limited exploration of the region where resources were last encountered, using

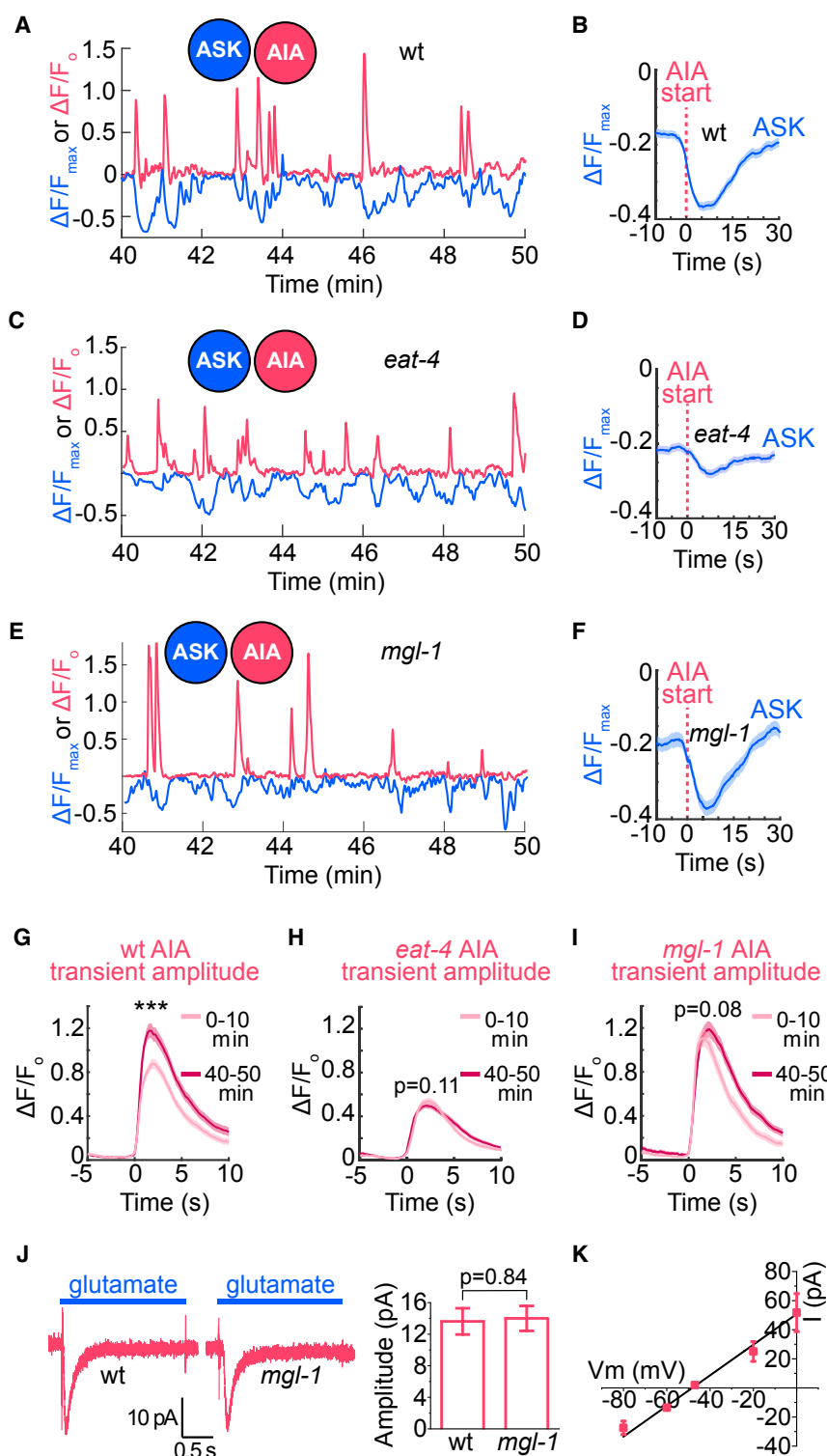


Figure 6. Glutamate Acts over Multiple Timescales to Regulate ASK-to-AIA Activity

(A, C, and E) Representative traces of (A) wild-type, (C) *eat-4* mutant, or (E) *mgl-1* mutant in buffer showing spontaneous calcium dynamics in simultaneously recorded ASK and AIA neurons expressing GCaMP5A, 40–50 min after food removal.

(B, D, and F) ASK traces aligned to the start of AIA transients in (B) wild-type animals ($n = 19$ animals, 248 transients), (D) *eat-4* mutants ($n = 21$ animals, 410 transients), and (F) *mgl-1* mutants ($n = 12$ animals, 89 transients). Data presented as mean $\pm$ SEM (shaded region). Data obtained from imaging ASK and AIA simultaneously 40–50 min after food removal.

(G–I) Aligned AIA-positive transients early (0–10 min) versus late (40–50 min) after food removal for (G) wild-type (0–10 min, $n = 210$; 40–50 min, $n = 255$; same data as in Figure 5F), (H) *eat-4* (0–10 min, $n = 179$; 40–50 min, $n = 250$), and (I) *mgl-1* (0–10 min, $n = 137$; 40–50 min, $n = 240$). p values for difference in amplitude calculated using Kolmogorov-Smirnov test. Data obtained from imaging AIA individually.

(J) Representative electrophysiological traces (left) and average amplitudes (right) of glutamate-induced currents in AIA at the holding potential of -60 mV in wild-type ($n = 9$) or *mgl-1* mutants ($n = 10$). p value calculated using Wilcoxon signed-rank test.

(K) Current-voltage relationship of glutamate-induced currents when AIA was voltage-clamped at different holding potentials. The red linear fitting curve intersects zero current at the chloride equilibrium potential ($E_{Cl^-} = -47$ mV, calculated from the recording solutions used). $n = 4$ –9 for each data point.

chemosensory and mechanosensory circuit modules that each have the ability to generate local search and a third contribution from an internal change in the reorientation circuit.

Together with previous work, our results suggest the following model for sustained local search behavior (Calhoun et al., 2015; Chalasani et al., 2007; Gray et al., 2005; Hills et al., 2004; Wakabayashi et al., 2004). Removal from food leads to the activation of multiple sensory neurons, which release glutamate onto targets including AIA and ADE. Glutamate acts through MGL-1, a slow G protein-coupled glutamate receptor, and in parallel on fast glutamate-gated channels in AIA and in other interneurons, to drive local search.

a set of motor programs—reversals, turns, and changes in locomotion speed—that are undirected, apparently internally generated, and relatively nonspecific, as they appear in many other spontaneous and evoked behaviors. Here, we define two parallel

Based on its similarity to inhibitory group II metabotropic receptors (Dillon et al., 2015), we suggest that MGL-1 suppresses AIA and ADE neurotransmitter release for the 10- to 20-min period corresponding to local search. The reduction of either AIA or ADE

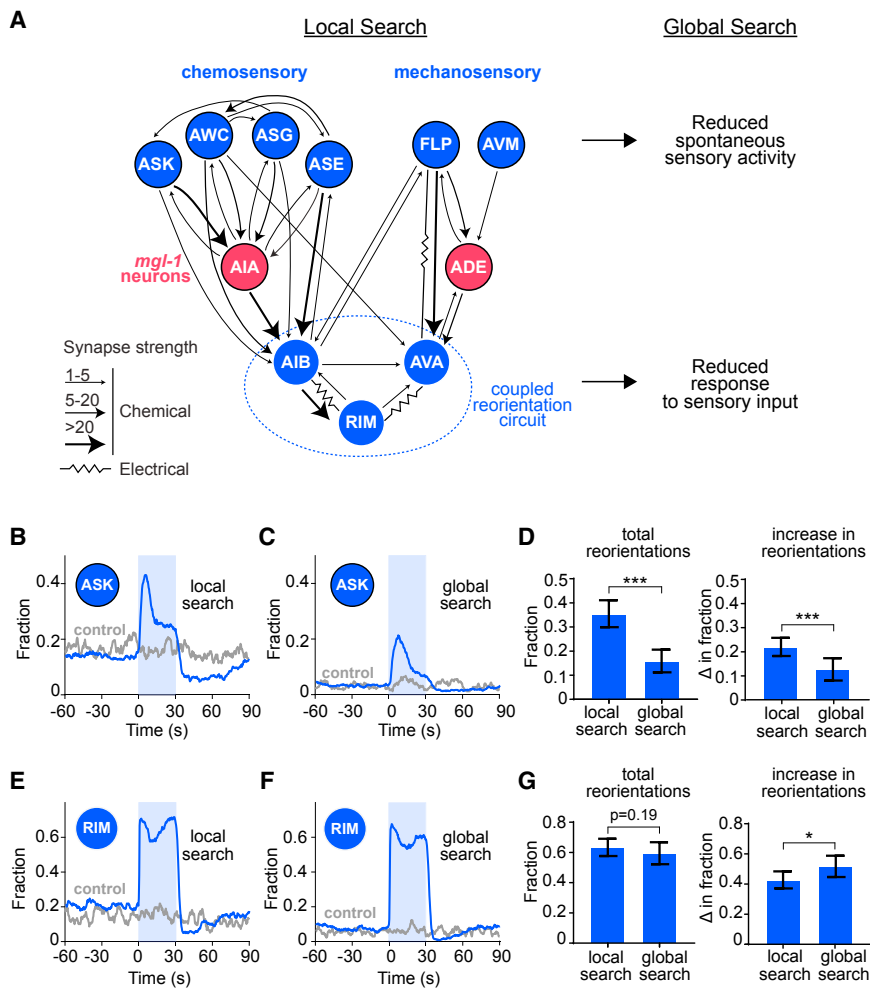


Figure 7. The Reorientation Circuit Reduces Its Response to Sensory Input over Time

(A) Synaptic map of parallel glutamatergic modules for local search. Neurons whose activity promotes local search behaviors are shown in blue; neurons that inhibit local search are shown in red. Arrows are weighted based on the number of chemical synapses from <http://www.wormweb.org>. Multimodal glutamatergic sensory neurons (blue, top) converge on downstream integrating neurons AIA and ADE (red, center), which express the inhibitory G protein-coupled receptor MGL-1. Transient suppression of AIA or ADE by MGL-1 is required for local search. Glutamate from the chemosensory neurons also activates fast glutamate-gated ion channels on AIA (Figure 6) and AIB (Chalasani et al., 2007). Sensory neurons and *mgl-1*-expressing neurons converge on AIB and AVA neurons (blue, bottom). AIB, RIM, and AVA neurons belong to a coupled network whose activity drives reversals and reorientations (Gordus et al., 2015; Kato et al., 2015). Right: the transition from local to global search is accompanied by changes in spontaneous sensory activity and by changes in the internal reorientation circuit that make it refractory to sensory input (B)–(G).

(B and C) Optogenetic activation of the ASK sensory neurons during (B) local search and (C) global search. Controls express ReaChR and are treated with light, but not pre-incubated with retinal. Data show the instantaneous fraction of animals performing a reorientation (ASK ReaChR, $n = 240$; controls, $n = 64$).

(D) Left: average fraction of animals performing reorientations during the first 10 s of ASK stimulation in local search and global search. Right: change in fraction of animals performing reorientations after stimulation of ASK, calculated by subtracting the pre-stimulation fraction (which is lower during global search) from the post-

stimulation fraction ($n = 15$ experiments). Data presented as mean $\pm$ SD. p values calculated using Wilcoxon rank-sum test *** $p < 0.001$.

(E and F) Optogenetic activation of the RIM integrating interneuron during (E) local search and (F) global search. Controls express ReaChR and are treated with light, but not pre-incubated with retinal. Data show the instantaneous fraction of animals performing a reorientation (RIM ReaChR, $n = 162$; controls, $n = 33$).

(G) Left: average fraction of animals performing reorientations during the first 10 s of RIM stimulation in local search and global search. Right: change in fraction of animals performing reorientations after stimulation of RIM, calculated by subtracting the pre-stimulation fraction (that is lower during global search) from the post-stimulation fraction ($n = 10$ experiments). Data presented as mean $\pm$ SD. p values calculated using Wilcoxon rank-sum test * $p < 0.05$.

In (B)–(G), 30-s green light stimuli were separated by 2-min recovery periods during the interval spanning from 2–12 min (local search) and 30–40 min (global search) after food removal. y axis, fraction of animals executing any time-dependent reorientation (Figures S1A–S1C), scored continuously.

release is sufficient to increase reversals and reorientations during this interval. At a circuit level, AIA and ADE neurotransmitter release may inhibit the reversal-promoting AIB and AVA neurons, respectively (Figure 7), consistent with the known antagonism between forward and reversal circuits (Roberts et al., 2016). After 10–20 min away from food, glutamatergic signaling from sensory neurons including ASK decreases, and at the same time, the reversal circuit becomes resistant to sensory input through additional mechanisms, resulting in global search behavior.

Parallel Sensory Circuits Generate Multimodal Control of Local Search

The local search circuit defined here is composed of two parallel modules: a set of glutamatergic chemosensory neurons that

synapse onto the AIA interneurons, and a set of glutamatergic mechanosensory neurons that synapse onto the ADE sensory-inter neurons. The chemosensory neurons detect food-related volatile odors (AWC) and amino acids (ASK), whereas the mechanosensory neurons detect touch and textures associated with food (Albeg et al., 2011; Sawin et al., 2000; Yemini et al., 2013). This circuit has several interesting features that shed light on multimodal sensory processing.

First, the chemosensory and mechanosensory modules that control local search are separate and parallel. There are no direct connections between the identified chemosensory and mechanosensory neurons, no shared connections onto the AIA or ADE integrating neurons, and no connections between AIA and ADE in the synaptic wiring diagram of *C. elegans* (White et al.,

1986). AIA and ADE primarily synapse onto AIB and AVA interneurons, respectively. AIB and AVA belong to a coupled neuronal group that is simultaneously activated during reversal behaviors (Gordus et al., 2015; Kato et al., 2015). The coupled activity of AIB and AVA is the first site of convergence between the chemosensory and mechanosensory modules. Modeling suggests that stochastic fluctuations between the coupled reversal circuit and a mutually inhibitory forward circuit may be sufficient for occasional reversals in the absence of overt sensory cues (Roberts et al., 2016); our results suggest that local search represents a change in the balance between these antagonistic assemblies.

Second, the AIA and ADE modules are each sufficient to generate full local search behavior: they are not additive. Suppressing the function of either AIA or ADE with MGL-1 or tetanus toxin permits local search. Similarly, glutamate release from either the chemosensory or the mechanosensory neurons can generate the full behavior. This redundant circuit organization may create a robust innate neuronal representation of food removal through multiple sensory modalities. Parallel processing of sensory and nutritional cues is characteristic of mammalian feeding and satiety circuits as well (Betley et al., 2013; Chen et al., 2015), suggesting that a redundant organization may be common in circuits involved in survival behaviors.

Fast and Slow Glutamate Receptors Cooperate in Local Search Behavior

Chemosensory neurons in the local search circuit regulate acute reorientations by activating excitatory and inhibitory ionotropic glutamate receptors on downstream interneurons (Hills et al., 2004; Chalasani et al., 2007). Here, we found that the same sensory neurons regulate sustained local search behavior by signaling through the G protein-coupled metabotropic glutamate receptor MGL-1.

Our results suggest that MGL-1 inhibits neurotransmitter release from AIA and ADE neurons, rather than affecting excitability. Animals with and without *mgl-1* have similar fast inhibitory glutamate-gated chloride currents in AIA, and similar anticorrelated ASK and AIA activity, suggesting that MGL-1 acts downstream of membrane potential. Based on sequence and pharmacology, *mgl-1* appears most similar to mammalian $G_{i/o}$ -coupled group II metabotropic receptors (Dillon et al., 2015), which inhibit neurotransmitter release (Dong and Ennis, 2018; Hayashi et al., 1993). The *C. elegans* G_o protein inhibits synaptic release from motor neurons (Miller et al., 1999; Nurrish et al., 1999; Vashlishan et al., 2008), suggesting that MGL-1 acting through G_o could similarly decrease synaptic release from AIA and ADE. In agreement with this possibility, inhibition of AIA or ADE synaptic release with tetanus toxin can fully substitute for *mgl-1* action in local search behavior, whereas decreasing excitability by hyperpolarizing AIA or ADE is less effective. The relevant AIA and ADE neurotransmitters might include acetylcholine (AIA), dopamine (ADE), and multiple neuropeptides (both neurons). Dopamine (*cat-2*) did not have an effect in this study, but it does have effects in other area-restricted search assays (Hills et al., 2004), and CEP dopaminergic neurons can regulate local search by integrating information about the size of a bacterial lawn over time (Calhoun et al., 2015). We did

identify the dopamine receptor DOP-2 in our initial mutant screen, providing one avenue for further study.

On a longer timescale, *mgl-1* coordinates a variety of behavioral and physiological responses to food depletion. *mgl-1* decreases pharyngeal pumping (feeding) after food removal (Dillon et al., 2015), promotes mobilization of fat stores downstream of transforming growth factor β (TGF- β) signaling (Greer et al., 2008), and modifies survival after starvation by regulating autophagy (Kang and Avery, 2009b). Its expression is regulated by AMPK and is dependent on satiety levels (Ahmadi and Roy, 2016). Identifying the cell-type-specific readouts of MGL-1 action will provide a better understanding of the short-term behavioral and longer-term physiological responses to changes in food availability.

What Dictates the Duration of Local Search?

Theoretical models have demonstrated that local search is the most successful strategy when animals have information about food availability, such as a recent food encounter, and a global search is most successful when animals need to locate random food sources with no prior information (Calhoun et al., 2014; Salvador et al., 2014). The circuits described here can encode the recent food encounter through multimodal sensory cues and thereby drive an active local search behavior that decays over time to a global search behavior. The slow reduction of ASK activity over many minutes and the associated rise in AIA activity represent neural correlates of the decaying food memory (our work and Skora et al., 2018). However, chronic tetanus toxin inhibition of AIA (and ADE) restores local search behavior even when sensory glutamate is absent. These results indicate that the glutamatergic memory is not essential, and that additional clocks can represent food history over time.

Although the nature of the additional clock is unknown, the reduced behavioral response to sensory activation during global search suggests an underlying change in a broader behavioral state. We speculate that neuropeptide co-transmitters from sensory neurons might mediate such a state change. Classical transmitters and neuropeptides cooperate in the rodent hypothalamus, where NPY co-transmitter release converts transient GABA-dependent synaptic effects into sustained feeding behavior (Chen et al., 2016). Most sensory neurons in the *C. elegans* local search circuit release FMRF neuropeptides, which are related to mammalian NPY peptides (Li and Kim, 2008), and several neuropeptides and G protein-coupled neuropeptide receptors affect local search behavior (Campbell et al., 2016; Chalasani et al., 2010). Neuropeptide co-transmitters that act antagonistically or synergistically with glutamate could explain the greater effects of sensory neuron ablation compared to glutamate knockouts (Wakabayashi et al., 2004; Gray et al., 2005), although other explanations are possible. An appealing aspect of this model is that neuropeptide release would be controlled by the same gradual decay in sensory activity as glutamate and MGL-1, but it is only one of many possibilities. Identifying the factors that shape internal states over slow timescales is the next step toward understanding how these essential behavioral states are sustained and terminated.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures and two tables and can be found with this article online at <https://doi.org/10.1016/j.neuron.2019.01.053>.

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AUTHOR CONTRIBUTIONS

A.L.-C. and C.I.B. designed experiments. A.L.-C. conducted experiments other than electrophysiological recordings. N.P. developed automated behavioral tracking code. A.L.-C. and A.S. generated the *eat-4* CRISPR glutamate knockout line. P.T.M. mapped the *mgl-1* deletion in CX17083. Q.L. designed and conducted electrophysiological experiments. A.L.-C. and C.I.B. analyzed and interpreted results and wrote the paper with input from all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, Peptides, and Recombinant Proteins		
Dow Corning Sylgard 184 Silicone Encapsulant Clear 0.5 kg Kit	Ellsworth Adhesives	Cat#184 SIL ELAST KIT 0.5KG
Dulbecco's Phosphate Buffered Saline	Sigma-Aldrich	Cat#D8537
Collagenase from Clostridium histolyticum, Type IV	Sigma-Aldrich	Cat#C5138
1,2-Bis(2-Aminophenoxy)ethane-N,N,N',N'-tetraacetic acid (BAPTA)	Sigma-Aldrich	Cat#A4926
Guanosine 5'-triphosphate sodium salt hydrate (NaGTP)	Sigma-Aldrich	Cat#G8877
Adenosine 5'-triphosphate disodium salt hydrate (Na ₂ ATP)	Sigma-Aldrich	Cat#A7699
Adenosine 3',5'-cyclic monophosphate (cAMP)	Sigma-Aldrich	Cat#A9501
Guanosine 2':3'-cyclic monophosphate sodium salt	Sigma-Aldrich	Cat#G8129
L-Glutamic acid monosodium salt hydrate	Sigma-Aldrich	Cat#G1626
Deposited Data		
Complete foraging assay data	Mendeley data	https://data.mendeley.com/datasets/prdtw5jgcd/1
Experimental Models: Organisms/Strains		
<i>mgl-1(ky1060) npr-9(tm1652) X</i>	This study	CX17083
<i>lgc-46(ok2949)</i>	This study	CX12722
<i>glr-1(n2461)</i>	CGC	KP4
<i>mgl-2(tm355)</i>	This study	CX12797
<i>acc-4(ok2371)</i>	This study	CX12724
<i>gab-1(tm3577)</i>	This study	CX12723
<i>glr-2(tm669)</i>	This study	CX12720
<i>lgc-48(gk964294)</i>	This study	CX16989
<i>npr-35(ok3258)</i>	This study	CX16990
<i>gar-2(ok520)</i>	This study	CX16986
<i>npr-4(tm1782)</i>	This study	CX13890
<i>nmr-1(ak4)</i>	Bargmann lab	VW4509
<i>seb-3(gk382193)</i>	This study	CX16403
<i>glr-3(ak57)</i>	Villu Maricq lab	VM1846
<i>acr-5(ok180)</i>	CGC	NC293
<i>avr-14(ad1305)</i>	This study	CX13755
<i>npr-5(ok1583)</i>	This study	CX14394
<i>glc-3(ok321)</i>	This study	CX16985
<i>ador-1(gk744003)</i>	This study	CX15563
<i>egl-6(n592)</i>	CGC	MT1222
<i>lgc-52(tm4596)</i>	This study	CX13889
<i>gar-1(ok755)</i>	This study	CX16983
<i>acc-2(ok2216)</i>	This study	CX16988
<i>npr-11(ok594)</i>	This study	CX8579
<i>glr-6(tm2729)</i>	This study	CX14904
<i>lgc-54(tm3448)</i>	This study	CX16982
<i>acc-1 (tm3268)</i>	This study	CX12721
<i>npr-14(ok2375)</i>	This study	CX16402
<i>ckr-2(tm2082)</i>	This study	CX15961
<i>dop-1(vs101)</i>	CGC	LX636

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
<i>dop-4(ok1321)</i>	This study	CX11751
<i>dop-3(ok295)</i>	CGC	BZ873
<i>gbb-1(tm1406)</i>	This study	CX15899
<i>lgc-38(gk177664)</i>	This study	CX15901
<i>lgc-37(tm0864)</i>	This study	CX15900
<i>lgc-53(tm2735)</i>	This study	CX11500
<i>ntr-1(tm2765)</i>	This study	CX14102
<i>mgl-3(tm1766)</i>	This study	CX13156
<i>lgc-47(ok2963)</i>	This study	CX16987
<i>dop-2(vs105)</i>	CGC	LX702
<i>acr-12(ok367)</i>	CGC	VC188
<i>pdf-1(ok3425)</i>	This study	CX14295
<i>npr-9(ky1030) X</i>	This study	CX1030
<i>mgl-1(ky1037) X</i>	This study	CX1037
<i>mgl-1(ky1060) npr-9(tm1652) X</i>	This study	CX17083
<i>mgl-1(ky1060) npr-9(tm1652) X; kyEx5968</i>	This study	CX17094
<i>mgl-1(ky1060) npr-9(tm1652) X; kyEx6019</i>	This study	CX17218
<i>mgl-1(ky1037) X; kyEx6178</i>	This study	CX17560
<i>mgl-1(ky1037) X; kyEx6056</i>	This study	CX17312
<i>mgl-1(ky1037) X; kyEx6056; kyEx6062</i>	This study	CX17345
<i>mgl-1(ky1037) X; kyEx6056; kyEx6064</i>	This study	CX17347
<i>mgl-1(ky1037) X; kyEx6056; kyEx6072</i>	This study	CX17354
<i>mgl-1(ky1037) X; kyEx6056; kyEx6076</i>	This study	CX17359
<i>mgl-1(ky1037) X; kyEx6056; kyEx6079</i>	This study	CX17362
<i>mgl-1(ky1037) X; kyEx6056; kyEx6066</i>	This study	CX17349
<i>mgl-1(ky1037) X; kyEx6056; kyEx6067</i>	This study	CX17350
<i>mgl-1(ky1037) X; kyEx6056; kyEx6073</i>	This study	CX17356
<i>mgl-1(ky1037) X; kyEx6056; kyEx6075</i>	This study	CX17358
<i>mgl-1(ky1037) X; kyEx6088</i>	This study	CX17393
<i>mgl-1(ky1037) X; kyEx6088; kyEx6089</i>	This study	CX17394
<i>mgl-1(ky1037) X; kyEx6098</i>	This study	CX17419
<i>mgl-1(ky1037) X; kyEx6101</i>	This study	CX17422
<i>mgl-1(ky1037) X; kyEx6098; kyEx6101</i>	This study	CX17457
<i>mgl-1(ky1037) X; kyEx6245</i>	This study	CX17738
<i>mgl-1(ky1037) X; kyEx6109</i>	This study	CX17441
<i>mgl-1(ky1037) X; kys698</i>	This study	CX17418
<i>mgl-1(ky1037) X; kyEx6246</i>	This study	CX17738
<i>kySi77 kySi76 III</i>	This study	CX17461
<i>kySi77 kySi76 III; kyEx6123</i>	This study	CX17471
<i>mgl-1(ky1037) X; kySi77 kySi76 III</i>	This study	CX17505
<i>kySi77 kySi76 III; kyEx6142</i>	This study	CX17524
<i>kySi77 kySi76 III; kyEx6150</i>	This study	CX17539
<i>kySi77 kySi76 III; kyEx6169</i>	This study	CX17551
<i>kySi77 kySi76 III; kyEx6153</i>	This study	CX17535
<i>kySi77 kySi76 III; kyEx6174</i>	This study	CX17556
<i>kySi77 kySi76 III; kyEx6157</i>	This study	CX17564
<i>kyEx6157</i>	This study	CX 17667
<i>kyEx6203; kyEx6202</i>	This study	CX17603

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
<i>kySi77 kySi76 III; kyEx6157; kyEx6101</i>	This study	CX17572
<i>kySi77 kySi76 III; kyEx6157; kyEx6098</i>	This study	CX17571
<i>kyEx6191</i>	This study	CX17590
<i>kyEx6115</i>	This study	CX17449
<i>mgl-1(ky1037); kyEx6115</i>	This study	CX17455
<i>kyEx6115; kyEx6191</i>	This study	CX17598
<i>mgl-1(ky1037) X; kyEx6115; kyEx6191</i>	This study	CX17614
<i>eat-4(ky5) III; kyEx6115; kyEx6191</i>	This study	CX17624
<i>kyEx6251</i>	This study	CX17746
<i>kyEx6236</i>	This study	CX17694
Recombinant DNA		
pALC01[FRT-let-858 3I-UTR-FRT donor plasmid]	This study	N/A
pALC02[mCherry + Cbr-unc-119 donor plasmid]	This study	N/A
pALC03 [combined plasmid vector]	This study	N/A
pALC04[npr-9 gRNA]	This study	N/A
pALC05[mgl-1 gRNA]	This study	N/A
pALC06 [mgl-1 first half::LOXinv(mgl-1 s half::sl2GFP)]	This study	N/A
pSF11[tag-168::nCre]	Bargmann Lab	N/A
pSF68[mod-1::nCre]	Bargmann Lab	N/A
pACL07[gcy-28d::nCre]	This study	N/A
pALC08[dat-1::nCre]	This study	N/A
pALC09[tph-1::nCre]	This study	N/A
pYX5[str-3::nCre]	Bargmann Lab	N/A
pXJ74[odr-2b::nCre]	This study	N/A
pALC10 [mgl-1 first half::LOXinv(mgl-1 s halfSTOP::sl2GFP)]	This study	N/A
pALC11[ttx-3intron7::TeTx::sl2GFP]	This study	N/A
pALC12[dat-1::TeTx::sl2GFP]	This study	N/A
pALC13[tph-1(short)::TeTx::sl2GFP]	This study	N/A
pCEC16[ttx-3intron7p::HisCl1::sl2::GFP]	Bargmann Lab	N/A
pALC14[dat-1::HisCl::sl2GFP]	This study	N/A
pALC15[tag-168::nFLP]	This study	N/A
pALC16[odr-1::nFLP]	This study	N/A
pALC17[sra-9::nFLP]	This study	N/A
pALC18[gcy-15::nFLP]	This study	N/A
pALC19[tax-4::nFLP]	This study	N/A
pALC20[mec-3::nFLP]	This study	N/A
pJL2[gcy-28 dp::GCaMP5 (D380Y)]	Bargmann Lab	N/A
pALC21[sra-9::GCaMP5A]	This study	N/A
pALC22[tax-4::inv(GtACR2::sl2GFP)]	This study	N/A
pALC23[mec-3::inv(GtACR2::sl2GFP)]	This study	N/A
pSF176[eat-4::nCre]	Bargmann Lab	N/A
pALC24[ttx-3intron7::ReaChR::sl2GFP]	This study	N/A
pALC25[sra-9::ReaChR::sl2GFP]	This study	N/A
pAS01[tdc-1::ReaChR]	This study	N/A
Software and Algorithms		
Foraging Analysis Software	This study	https://github.com/navinpokala/BargmannWormTracker
MetaMorph (Version 7.7.9.0)	BioVision Technologies	https://www.biovis.com/metamorph.html

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Streampix 6	Norpix	https://www.norpix.com/products/streampix/streampix.php
PatchMaster (v2X90)	HEKA	http://www.heka.com
FitMaster (v2x90.1)	HEKA	http://www.heka.com
Other		
PL-D7715 15 MP Camera	PixLink	N/A
Axioskop 2	Zeiss	N/A
iXon3 DU-897 EMCCD camera	Andor	N/A
SOLIS-525B High Power LED	Thor Labs	N/A
3M Vetbond tissue adhesive	Fisher Scientific	Cat#NC0304169
Abbott Laboratories GLUture Topical Adhesive	Amazon	Cat#B00IJVQVO6
Diamond dissecting blade (Type M-DL)	EMS	Cat#72029-L
Fisherbrand Microscope Cover Glass	Fisher Scientific	Cat#12-545-101 22CIR.-1
Borosilicate glass pipettes	Sutter Instruments	Cat#BF100-58-10
EPC-10 USB amplifier	HEKA	N/A
SZX16 Stereomicroscope	Olympus	N/A
Axio Examiner upright microscope	Carl Zeiss	N/A
PicoSpritzer III	Parker Hannifin	N/A
PatchStar Micromanipulator	Scientifica	N/A
Lumencor Spectra X light engine	Lumencor	N/A

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Dr. Cornelia Bargmann (cori@rockefeller.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS***C. elegans* strains**

All strains are listed in [Tables S1](#) and [S2](#). Existing mutant strains used for the candidate genetic screen are listed in [Table S1](#). New CRISPR-generated strains and transgenic strains are listed in [Table S2](#).

Nematode Culture

Animals were grown at room temperature (21–22°C) on nematode growth media (NGM) plates seeded with *E. coli* OP50 bacteria ([Brenner, 1974](#)). Wild-type animals were *C. elegans* Bristol strain N2. All mutant strains tested were backcrossed to wild-type animals to reduce unlinked background mutations. Transgenic strains were generated by microinjection of a transgene derived from wild-type N2 DNA, a fluorescent co-injection marker (myo-2::mCherry, myo-3::mCherry, elt-2::nls-GFP, elt-2::mCherry), and empty pSM vector to reach a final DNA concentration of 100 ng/μL.

Transgenic and mutant strains were always compared to matched controls tested in parallel on the same days. Each experiment included 12–18 animals. For the candidate genetic screen, each mutant strain was tested in six different experiments done on two different days. All other strains were tested in 4–8 experiments done on at least two different days.

METHOD DETAILS**Generation of CRISPR/Cas9 mutant strains**

CX1030 *npr-9(ky1030)* X. *ky1030* is a CRISPR/Cas9 induced single nucleotide deletion in the second exon of *npr-9*. The resulting sequence is TGGTAATGCTCTGGTGGTGAT (deleted nucleotide is underlined). To generate the strain, we used a co-CRISPR protocol ([Aribere et al., 2014](#)). Young hermaphrodites were injected with a mix of plasmids encoding Cas9, a gRNA targeting *rol-6*, a gRNA targeting *npr-9*, and a ssDNA repair template that induces a dominant *rol-6(su1006)* mutation. F1 animals were isolated to individual plates based on their roller phenotype, allowed to lay eggs, and then screened for a target mutation by Sanger sequencing. F2 animals were isolated to find homozygotes for the target mutation. After inducing the target mutation, animals were backcrossed to wild-type twice.

CX1037 *mgl-1(ky1037)* X. *ky1037* is a CRISPR/Cas9 induced indel that results in a frameshift before the first transmembrane domain of *mgl-1*. Deletion is (G), insertion is (TTGTGTGGTTGTGTGGTTGTGGTTGTGTGTTGT). Resulting sequence is TGGAGCA ACGTTGTGTGGTTGTGTGGTTGTGGTTGTGTGTTGTGGTGGTTCT (insertion is underlined). We used the co-CRISPR protocol as described above (Arribere et al., 2014).

pJA42 was a gift from Andrew Fire (Addgene plasmid # 59930).

CRISPR/Cas9 editing of the endogenous *eat-4* (VGLUT1) locus

To generate a strain that would allow us to knock out endogenous glutamate release in a cell-specific manner, we performed two successive edits on the endogenous *eat-4* locus.

First, we used the CRISPR/Cas9 SapTrap method (Schwartz and Jorgensen, 2016) to insert *let-858 3'-UTR::FRT::mCherry* immediately after the endogenous *eat-4* stop codon. SapTrap is a modular plasmid assembly approach that produces a single plasmid vector containing both a gRNA transcript and a repair template. We mutated some of the modular components used to assemble the repair template as follows: for pMLS279 (*FRT-let-858 3'-UTR-FRT*) we deleted the 5' FRT site (new plasmid is pALC01), and for pMLS291 (mCherry with sytron embedded inverted floxed *Cbr-unc-119*) we added a stop codon at the end of the mCherry coding region (new plasmid is pALC02). We cloned these two edited plasmids along with 60 bp homology arms and the gRNA into the pMLS256 destination vector, producing a combined plasmid vector (pALC03). Young *unc-119(ed3)* hermaphrodites were injected with the combined plasmid vector, a Cas9 expression vector, and fluorescent co-injection markers. F1 animals were picked 10 days after injection based on rescue of the *unc-119* phenotype, and the insertion was confirmed by PCR and Sanger sequencing. These edited animals were then injected with *peft-3::nCre* (pDD104) to excise the sytron embedded, inverted floxed *Cbr-unc-119*. F1 animals were selected based on the reappearance of the *unc-119* phenotype. This line was backcrossed once to wild-type animals to remove the *unc-119(ed3)* mutation, and an additional three times to remove any off-target CRISPR mutations.

Second, we used the co-CRISPR method described above (Arribere et al., 2014) to insert an FRT sequence (GAAGTTCCTA TTCTCTAGAAAGTATAGGAAGTTC) immediately before the start codon of *eat-4*. We injected the mix described above with an additional ssDNA repair template consisting of the FRT sequence with 35 bp homology arms on each side. The resulting edited genomic sequence is: TCATCATCATTTTCAGAAACCGAAGTTCCTATTCTCTAGAAAGTATAGGAAGTTCATGTCGTCATGGAACGAaGC (FRT insertion is underlined; *eat-4* locus is in italics; bolded ATG is the start codon of *eat-4*; lowercase 'a' is a silent mutation induced to remove PAM site after successful homologous recombination).

Nuclear localized flippase (nFLP) was subcloned from pMLS262 (snt-1::2xNLS-FLP-D5) into pSM using Gibson assembly (New England Biolabs). A plasmid containing nFLP under cell specific promoters was injected into the *eat-4* edited strain (CX17461) by standard gonadal microinjection. Successful glutamate knockout in the target cells was confirmed by mCherry expression which was visualized using a Zeiss Axio Imager.Z1 Apotome microscope with a 40x objective. For all FLP-dependent glutamate knockout lines in Figure 4, 20/20 animals examined had mCherry expression bilaterally in the target cells.

For all glutamate knockout experiments (Figure 4), 'control' animals are the *eat-4* edited strain, and '*mgl-1*' are *mgl-1(ky1037)* *eat-4* edited strain.

pMLS279 (Addgene plasmid # 73729), pMLS291 (Addgene plasmid # 73724), pMLS256 (Addgene plasmid # 73715), and pMLS262 (Addgene plasmid # 73718) were gifts from Erik Jorgensen.

Identification of *mgl-1(ky1060)* allele

We prepared genomic DNA from strain CX17083 using standard phenol/chloroform extraction protocols. Sequencing was conducted at the Rockefeller High-Throughput Sequencing Facility, where a DNA library was prepared using TruSeq Nano DNA kit (Illumina) and sequenced in an Illumina HiSeq 2500 System.

Deletions were identified using modifications to a previously described approach (McGrath et al., 2011). Candidate deletions were first identified using 'chimeric' reads as identified by bwa (i.e., reads that included the SA tag) where each alignment mapped to the same chromosome and strand (Li and Durbin, 2010). These reads were used to infer the breakpoints and insertion sequence of a candidate deletion. Candidate deletions that were also present in N2 sequenced controls were excluded as likely errors in the reference sequence. Each candidate deletion was then genotyped by collecting all the reads with primary alignments that fell within 10 bp of the candidate deletion and a mapping quality score greater than 10. These reads were realigned to both the reference and the candidate deletion sequence using a striped Smith Waterman Alignment implemented in the scikit-bio Python library (<http://scikit-bio.org/>). This analysis identified a homozygous 225 bp deletion in the *mgl-1* gene that was verified using Sanger sequencing.

The identified *mgl-1* deletion is: ATCTGCTCAACGACCAAGATTCATATCTCCCATCTCTCAGgtgagctccggtgacaagccaacggaagta cactatttatagGTTGTCATGACTGCAATGCTAGCCGGAGTACAATTGATCGGAAGTCTTATTTGGCTGTCAGTAGTGCCACCAAGgtaaa ttggctattatgaagtgtctgtagtaatttttagGTTGGAGACACCACTACCCCAACAGGGACCAAGGTGGTTTAACTTGTAATGTTCTCTGACCA TCACCTTTTGTATTC (deletion in italics, exons in uppercase, introns in lowercase)

Intersectional Cre/Lox *mgl-1* rescue

To rescue *mgl-1* in subsets of its endogenous expression pattern, we employed an intersectional transgene strategy depicted in Figure 3A. The inactive *mgl-1* genomic fragment plasmid was constructed using pSM-inv[sl2-GFP] as the backbone (Flavell et al., 2013). A portion of a *mgl-1* rescue genomic fragment sequence was cloned into pSM-inv[sl2-GFP] in its correct orientation (obtained using

forward primer: 5'- GTAAGGTATGTTTTATTTTCCAAC -3' and reverse primer: 5'- TAGAACAGACAAACATATTTGAC -3' and cloned in before the first Lox2272 and LoxP sites). The remaining *mgl-1* fragment sequence was cloned into pSM-inv[sl2-GFP] in an inverted orientation (obtained using forward primer: 5'- TCATAAGAAAGTATCGTGAGC -3' and reverse primer: 5'- CATTATGGCGTATGATGGG -3', cloned in after the inverted sl2-GFP and before the inverted LoxP and Lox2272 sites). The cloning was done via Gibson assembly (New England Biolabs).

To rescue *mgl-1* in subsets of its endogenous expression pattern, we first injected the partially inverted plasmid described above into *mgl-1(ky1037)* animals using standard microinjection protocols. The resulting line is labeled 'no Cre' in Figures 3B, 3C, and S4A–S4C. Subsequently, we injected plasmids expressing nls-Cre under cell specific promoters into this 'no Cre' line. This led to Cre-mediated inversion and reconstitution of the *mgl-1* rescue fragment in subsets of cells, which was confirmed by GFP expression visualized using a Zeiss Axio Imager.Z1 Apotome microscope with a 40x objective.

Foraging assay and quantification

Bacterial food lawns were made by seeding NGM plates with a thin uniform OP50 lawn (OD~0.5) 16 hours before the assay. The lawn covered the entire plate to eliminate effects of animals exploring the lawn edge (Calhoun et al., 2015), and contained a filter paper barrier soaked in 20 mM CuCl₂ that prevented the animals from leaving a 5 cm x 5 cm region on food. On the assay day, 10–15 adult hermaphrodites were first transferred for 45 min to this standard food lawn. To study off-food foraging, animals were transferred from the standard food plate to an unseeded NGM plate, allowed to crawl at least five body lengths to clean off excess food, and transferred to the assay plate which consisted of a large NGM plate with a circular filter paper barrier (~80 cm²) soaked in 20 mM CuCl₂ to restrict animals to the recorded area. Their behavior was recorded for 45 min, starting four minutes after the initial food removal, using a 15 MP PL-D7715 CMOS video camera (Pixelink). Frames were acquired at 3 fps using Streampix software (Norpix). Individual worm trajectories were analyzed using custom MATLAB (Mathworks) software, as previously described (Pokala et al., 2014). We were able to track the behavior of some individuals for the entire 45 min of food. When collisions occurred, however, the data points around the collision were excluded (to ignore reorientations associated with collisions) and sometimes we were unable to link tracks after the collision, resulting in separate tracks before and after collisions. These track fragments were included in average quantifications.

To quantify reorientation frequencies, we counted the number of reorientations that each animal performed in 2-minute time windows and divided it by the number of animals tracked in that time window. We only counted animals that we were able to track in the entire 2-min time window. The plots show the mean number of reorientations in 2-min time windows. In all figures except Figure S1, we quantified the number of time-dependent reorientations only (reorientations shown in Figures S1A–S1C).

To calculate Mean Square Displacement (MSD) (Figure 1B), we used the centroid x-y position of each animal. MSD was defined as the square of the distance traveled in one minute. For each animal, we first we calculated the MSD for each frame by measuring the squared distance traveled from 30 s before the frame to 30 s after the frame. We then calculated the average MSD in two-minute time windows for each animal (the average MSD in 360 frames = 2 min).

For the food concentration experiments (Figure 1D), the usual OP50 culture (OD~0.5) was diluted 10x or 100x in LB before seeding on NGM plates and growing for 16 hours overnight. The animals were placed on these plates for 45 min before the assay began. The off-food experiment was the same as described above.

Histamine silencing experiments

(Pokala et al., 2014) (<https://www.protocols.io/view/histamine-hiscl1-system-for-c-elegans-mumc6u6>)

Neurons expressing the histamine-gated chloride channel (HisCl1) were silenced by adding histamine to NGM assay plates. 1 M histamine-dihydrochloride (Sigma-Aldrich) in deionized water was filtered and stored at –20°C. NGM histamine plates were made by adding 1M histamine to NGM solution (50–55°C) to a final concentration of 50 mM histamine. Plates were stored at 4°C and used within a week. Controls included: (1) animals expressing the HisCl1 tested on normal plates and (2) animals with no transgene tested on histamine NGM plates.

Calcium imaging

Transgenic animals expressing GCaMP5A (Akerboom et al., 2012) in ASK (*sra-9::GCaMP5A*), in AIA (*gcy-28d::GCaMP5A*), or in both neurons (crossing the two single neuron GCaMP5A lines), were generated in a wild-type background and crossed into *mgl-1* or *eat-4* if indicated. Adult animals were first transferred to a standard food lawn for 45 min prior to imaging. Animals were then removed from the lawn, washed in NGM buffer to remove any food, and loaded into a custom PDMS microfluidic chamber in NGM buffer, where they were physically restrained but not paralyzed (Chronis et al., 2007). Animals received a constant flow of NGM buffer through the experiment. Imaging began 6 min after removal from food (time = 0 min). Animals were imaged for 10 min initially (0–10 min), left in the device for 30 min with the light off, and imaged again for another 10 min (40–50 min).

Electrophysiological recording of AIA

Electrophysiological recording was performed largely as previously described (Liu et al., 2018). Briefly, an adult animal was immobilized on a Sylgard-coated (Sylgard 184, Dow Corning) glass coverslip in a small drop of DPBS (D8537; Sigma) by applying a cyanoacrylate adhesive (Vetbond tissue adhesive; 3M) along the dorsal side of the head region. A puncture in the cuticle away from the head was made to relieve hydrostatic pressure. A small longitudinal incision was then made using a diamond dissecting blade

(Type M-DL 72029-L; EMS) between the two pharyngeal bulbs along the glue line. The cuticle flap was folded back and glued to the coverslip with GLUture Topical Adhesive (Abbott Laboratories), exposing the nerve ring. The coverslip with the dissected preparation was then placed in a custom-made open recording chamber (~1.5 mL volume) and treated with 1 mg/ml collagenase (type IV; Sigma) for ~10 s by hand pipetting. The recording chamber was subsequently perfused with ~10 mL of the desired extracellular solution with a custom-made gravity-feed perfusion system. All experiments were performed with the bath at room temperature. An upright microscope (Axio Examiner; Carl Zeiss, Inc.) equipped with a 40 X water immersion lens and 16 X eyepieces was used for viewing the preparation. AIA was identified with the GCaMP fluorescent marker. Borosilicate glass pipettes (BF100-58-10; Sutter Instruments) with resistance (RE) = 10–15 M Ω pre-pulled using a laser pipette puller (P-2000; Sutter Instruments) were back-filled with desired intracellular solution to use as electrodes. A motorized micromanipulator (PatchStar Micromanipulator; Scientifica) was used to control the electrodes. Experiments were conducted on an EPC-10 amplifier (EPC-10 USB; Heka) using PatchMaster software (Heka). Two-component capacitive compensation was optimized at rest, and series resistance was compensated to 50%. Analog data were filtered at 2 kHz and digitized at 10 kHz. As the quality control, only those patch clamps with seal resistance above 1 G Ω and uncompensated series resistance below 100 M Ω were accepted for further analysis. The pipette solution was (all concentrations in mM) [K-gluconate 115; KCl 15; KOH 10; MgCl₂ 5; CaCl₂ 0.1; Na₂ATP 5; NaGTP 0.5; Na-cGMP 0.5; cAMP 0.5; BAPTA 1; HEPES 10; Sucrose 50] (pH was adjusted with KOH to 7.2, osmolarity 320–330 mOsm). The extracellular solution was [NaCl 140; NaOH 5; KCl 5; CaCl₂ 2; MgCl₂ 5; Sucrose 15; HEPES 15; Dextrose 25] (pH was adjusted with NaOH to 7.3, osmolarity 330–340 mOsm). Liquid junction potential was calculated and corrected before recording. For glutamate application, L-Glutamic acid (Sigma) dissolved in the extracellular solution to 1 mM final concentration and filled into a glass pipette with RE = 1–3 M Ω was spritzed with PicoSpritzer III (Parker Hannifin Co.) at ~2.5 psi onto the cell being recorded. The bath was under active perfusion during glutamate application. Multiple trials with at least 10 s intertrial intervals were conducted on the same cell at different holding potentials when possible.

Optogenetic neuronal manipulations

To silence neurons optogenetically we expressed the light-gated anion channel, *Guillardia theta* anion channel rhodopsin 2 (*GtACR2*). A codon-optimized *GtACR2* was synthesized based on the *GtACR2* ORF (Govorunova et al., 2015) and cloned into the pSM vector backbone via Gibson Assembly (New England Biolabs). To activate ASK and RIM we expressed the red shifted channelrhodopsin variant ReaChR under *sra-9* and *tdc-1* promoters, respectively.

To express *GtACR2* selectively in glutamatergic sensory neurons that synapse onto AIA and ADE (the sensory neurons in which glutamate knockout leads to local search defect, Figure 4), we employed an inverted Cre-Lox recombination strategy. A floxed *GtACR2* cDNA was placed in an inverted orientation under the *tax-4* (chemosensory subset) and *mec-3* (mechanosensory subset) promoters, and subsequently activated in the relevant cells by Cre expression under the glutamatergic *eat-4* promoter.

16 hours before the optogenetic experiments, L4 animals expressing *GtACR2* or ReaChR were picked to NGM plates seeded with OP50 containing 12.5 μ M all-trans retinal (Sigma). Control animals, which also expressed *GtACR2* or ReaChR, were transferred to NGM plates seeded with OP50 but no retinal. On the experimental day, 15–20 animals were first transferred to a standard food lawn (with or without retinal) for 45 min, and then transferred to the large off-food assay plate (~80 cm²) where their behavior was recorded as described above. During video recording, green light (~45 μ W/mm²) was delivered at 10 Hz (50% duty cycle) using a Solis High Power LED (Thorlabs) controlled with custom MATLAB software.

For the sensory silencing experiments, we delivered four 30 s light pulses with 2 min between each pulse. The pulses began 2 min after initiating recording. To analyze the effect of the neuronal manipulation, we calculated the fraction of animals reorienting in each frame. We aligned all pulses and averaged the behavior for ten different experiments (40 light pulses).

For the ASK and RIM stimulation experiments we delivered four 30 s light pulses with 2 min between each pulse. The pulses began 2 min after initiating recording. These first four pulses correspond to the local search stimulations. We then waited 28 min and delivered four more 30 s light pulses with 2 min between each pulse. These last four pulses correspond to the global search stimulations.

QUANTIFICATION AND STATISTICAL ANALYSIS

Difference in local search behavior and datasets

Differences in local search behavior between groups was assessed by counting the number of reorientations that each animal in each group performed 2–12 min after food removal. Statistical significance was assessed using a Wilcoxon rank-sum test with Bonferroni correction.

The Wilcoxon signed-rank test was used to compare matched data (imaging experiments, same animal early versus late). The Kolmogorov-Smirnov test was used for comparing cumulative probability distributions. Statistical details for each experiment can be found in the figure legends.

Calcium imaging and data analysis

Calcium fluorescent signals were acquired at 5 frames per second through a 40x objective on an upright Axioskop 2 microscope (Zeiss) with an iXon3 DU-897 EMCCD camera (Andor), using Metamorph Software (Molecular Devices). Custom ImageJ and MATLAB software were used to track, quantify, and analyze neuronal activity. ASK imaging data were bleach-corrected. AIA did not show significant bleaching.

For ASK, the maximum baseline fluorescence in 1-minute intervals, $F_{\max}$ (95th percentile value), was used to calculate the change in fluorescence for each frame in the interval ($\Delta F = F - F_{\max}$), and subsequently normalized to $F_{\max}$ ($\Delta F / F_{\max}$). Changes in fluorescence observed were spontaneous decreases from the ASK baseline fluorescence (negative transients). To characterize negative transients, we wrote custom MATLAB software where we defined parameters to identify transients (start/end of transients = $-0.1 \Delta F / F_{\max}$, minimum transient amplitude = $-0.3 \Delta F / F_{\max}$, minimum transient duration = 3 s). Additionally, we used the MATLAB function *findpeaks* to identify complex transients which consist of multiple local minima within one transient (arrowheads in Figure 5A). Parameters were optimized to match manually assigned negative transients. For transient number and amplitude quantifications we measured all local minima within a complex transient. For transient duration quantifications we calculated the duration of the entire complex transient. For *eat-4* analysis, we used the same parameters used in wild-type data.

For AIA, the minimum baseline fluorescence, F_0 , was calculated using the MATLAB function *statelevels* which divides the data into two histograms and computes the mode of the lower histogram. F_0 was used to calculate the change in fluorescence for each frame ($\Delta F = F - F_0$), and subsequently normalized to F_0 ($\Delta F / F_0$). AIA shows a low baseline fluorescence with spontaneous positive transients. To characterize positive transients, we used the same procedure described above (parameters: start/end of transients = $0.1 \Delta F / F_0$, minimum transient amplitude = $0.3 \Delta F / F_0$, minimum transient duration = 2 s). Parameters were optimized to match manually assigned transients. For transient number and amplitude quantifications we measured all local maxima within a complex transient. For transient duration quantifications we calculated the duration of the entire complex transient. For *mgI-1* and *eat-4* analysis, we used parameters used in wild-type data.

To align ASK traces to AIA positive transient onset in the simultaneous imaging experiments (Figure 6), we first identified the start time of each AIA transient, and then extracted ASK frames from 10 s (50 frames) before the start time until 30 s after the start time. We then averaged all the extracted ASK frames.

To calculate correlations, we first z-scored each trace and then computed the autocorrelation, or the cross-correlation between ASK and matching lagged AIA traces, using the MATLAB function *xcorr*.

DATA AND SOFTWARE AVAILABILITY

Raw data for all foraging experiments have been deposited into Mendeley (<https://data.mendeley.com/datasets/prdtw5jgcd/1>).

Behavioral tracking MATLAB software has been deposited into GitHub (<https://github.com/navinpokala/BargmannWormTracker>).